

Mid-Term Project

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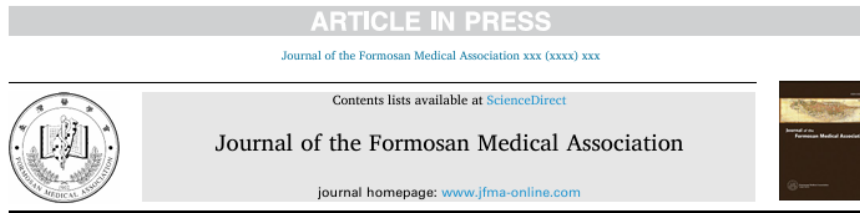
Online Instruction Folder

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Screening for Sarcopenia

- Refer to this article



Evaluating the role of toe grip strength and anthropometric measures in diagnosing sarcopenia: A predictive model for middle-aged and older adults in Taiwan

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ABSTRACT

Background: Sarcopenia, characterized by the loss of muscle mass and strength, is a major health concern among older adults, contributing to disability and mortality. Early identification is crucial for timely intervention. This study evaluated the role of toe grip strength (TGS) for diagnosing sarcopenia and developed a simplified diagnostic algorithm that reduces reliance on tools such as bioelectrical impedance analysis and handgrip strength.

Methods: A total of 771 participants (591 women, 180 men), aged 50 and above, were enrolled from a health management center in Taiwan. Participants underwent measurements of calf circumference, handgrip strength, TGS, and body composition. Sarcopenia was defined by the EWGSOP2 criteria: low muscle strength and reduced appendicular skeletal muscle mass index. Logistic regression models assessed the predictive power of age, TGS, and anthropometric measurements for sarcopenia diagnosis.

Results: Sarcopenia was identified in 18.3 % of participants, with a higher percentage in women (18.8 %) than in men (16.7 %). In women, age, height, and calf circumference were strong predictors of sarcopenia, and when combined, resulted in a high AUROC of 0.885. TGS, used alone, had a moderate AUROC of 0.736 and did not significantly improve discrimination when added to the model. In men, TGS was a stronger predictor (AUROC of 0.825), but adding it to the model did not significantly improve diagnostic performance.

Conclusion: Age, height, and calf circumference are effective in identifying sarcopenia in women. TGS may provide insights into mobility but does not enhance sarcopenia diagnosis. Future research should validate these findings and explore novel diagnostic tools.

https://drive.google.com/file/d/1gCJQBI0sSrmP12pv4AFKOF2oF9gQQ6O/view?usp=drive_link



The dataset for midterm practice

- Features
 - Age, calf circumference ...
- Targets
 - Muscle power (handgrip strength)
 - Muscle mass (ASML)
 - Sarcopenia
 - Low muscle strength: handgrip strength <28 kg for men and <18 kg for women.
 - Reduced skeletal muscle mass index: ASMI ≤ 7.0 kg/m² for men and ≤ 5.7 kg/m² for women, measured using bioelectrical impedance analysis (BIA)
- Dataset here:
 - CSV
 - https://drive.google.com/file/d/1KFgNe3pBjZJvC9Uhn4DYORkP0DeGgTOU/view?usp=drive_link
 - Excel
 - https://docs.google.com/spreadsheets/d/16a39Y3Em1-rtNVaigpj7L5T8pSf8zXgL/edit?usp=drive_link&ouid=110407566279501444493&rtpof=true&sd=true
 - SPSS
 - https://drive.google.com/file/d/1LtQk6SgyrNnDvsOlseFjZp--jjp7lGGG/view?usp=drive_link



CSV



Excel



SPSS

The dataset for practice

- Women who participated in the sarcopenia screening program
- 591 subjects
- 10 predicting variables (features)
- 2 outcome variables (muscle power and muscle mass)
 - (Low) **muscle power** (classification or **regression**)
 - (Low) **muscle mass** (classification or **regression**)
- **Main Target: Sarcopenia**
 - **Binary classification**
- Codebook here:
 - https://docs.google.com/document/d/14mw0VpJWeR9zs7fd-p3NDu-TxSO1LSXY/edit?usp=drive_link&oid=110407566279501444493&rtpof=true&sd=true



Variables	Label	Value	Role	Quote
seq_id	Individual id number	Numeric	No, just an id number	Not used in model
sex	Sex	1: men 0: women	feature	Only women included in the study
age	Age (years)	Numeric	feature	
dominant_hand	Dominant hand	1: right 0, left	feature	
dominant_leg	Dominant leg	1: right 0, left	feature	
calf_circ	Calf circumference (cm) of the dominant leg	Numeric	feature	
weight	Weight (kg)	Numeric	feature	
height	Height (cm)	Numeric	feature	
BMI	Body mass index, defined as weight (kg) divided by square of height (kg/m ²)	Numeric	feature	Probably no need to use this variable once weight and height are included
function_survey	A questionnaire survey about functional status	Numeric (ranging from 0 to 7)	feature	This variable is ordinal. But it is OK to treat it as continuous.
new_test	A new test that has not been evaluated yet and measured some kind of power (kg)	Numeric	feature	A new test that is not used in the diagnostic criteria for sarcopenia
muscle_power	Muscle power measured from dynamometer, indicating handgrip strength (kg))	Numeric	Target (continuous)	A target for regression
muscle_mass	Muscle mass in terms of appendicular skeletal muscle power (ASML, kg/m ²) measured by bioelectrical impedance analysis (BIA)	Numeric	Target (continuous)	A target for regression
sarcopenia	Sarcopenia was defined based on the criteria established by EWGSOP2. Participants were classified as having sarcopenia if they met both of the following conditions: 1. Low muscle strength: handgrip strength <28 kg for men and <18 kg for women. 2. Reduced skeletal muscle mass index: ASMI ≤7.0 kg/m ² for men and ≤5.7 kg/m ² for women, measured using BIA.	1: Sarcopneia 0: No sarcopenia	Main Target (binary)	The main target, combining both low muscle power and low muscle mass, is a binary outcome for classification.

Analytic Issues

- Descriptive statistics
- Continuous data vs. categorical data
- Missing data (very few):
 - Listwise deletion
 - Categorization
 - Imputation
- Only the 433 women aged 65 years old and above
- Two analytic strategies:
 1. Selecting those who are aged (> 65 y/o)
 2. Do twice:
 - 1) All women ($n = 592$)
 - 2) Those who are aged ($n = 433$)

Regression vs. Classification

Regression

- To predict how much the muscle power and muscle power would be
- Statistical methods
 - Linear regression
- Machine learning
 - Ridge regression
 - Lasso regression
 - Tree-based ensemble methods ...

Classification

- To predict whether a woman would have sarcopenia
- Statistical methods
 - Logistic regression
- Machine learning
 - Multiple level neural networks
 - Support vector machine (SVM)
 - Bayesian analysis
 - Random forest
 - Xgboost ...

Requirement

- Statistical models
 - Build a linear regression model for predicting muscle power
 - Build a linear regression model for predicting muscle mass
 - Build a logistic regression model for predicting the presence of sarcopenia

Requirement

- Machine learning models
 - Build a ML model for predicting muscle power and muscle mass
 - Choose one appropriate model, either ridge regression, Lasso regression, tree-based ensemble methods or others
 - Build a ML model for predicting the presence of sarcopenia
 - Choose one appropriate model, either NN, SVM, Bayesian analysis, Random forest , Xgboost, or others

Requirement

- Evaluation
 - Regression model
 - Mean absolute error (RAE), root mean square error (RMSE), R^2
 - Whatever appropriate
 - Classification model
 - Receiver operating characteristic (ROC) curve
- Also work on
 - Feature relative importance
 - Developing parsimonious model